Digital Note-Taking for Writing

Joanna Pitura

Abstract Note-taking is prevalent in academia—it is the basis of scholarly work, i.e. searching for information, collecting and reading literature, writing and collaborating, referred to as a "primitive" that assists these information activities (e.g., Palmer, C. L., Teffeau, L. C., & Pirmann, C. M. (2009). Scholarly information practices in the online environment: Themes from the literature and implications for library service development. OCLC Research. https://accesson.kisti.re.kr/upload2/i_report/1239602399570.pdf). Researchers and higher education students take notes throughout the inquiry cycle, i.e. while designing research, collecting data, analysing data, and writing the report. In addition, with written assignments being a considerable part of student academic work, notes are taken in the writing process, from generating ideas for writing tasks, through text planning and drafting to its editing. As this process may be challenging, digital note-taking has the potential to facilitate writing in academic contexts (Matysek, A., & Tomaszczyk, J. (2020). Digital wisdom in research work. Zagadnienia Informacji Naukowej—Studia Informacyjne, 58(2A(116A)), 98–113. https://doi.org/10.36702/zin.705). Yet, despite the availability of literature concerning formal requirements of writing, such as style, structure, referencing, etc., relatively little literature deals with the note-taking activity that assists academic writing, and even less with digital note-taking. In order to bridge this gap, this chapter focuses on the note-taking activity and shows how digital tools can support note takers in the academic writing context.

Keywords Digital note-taking · Academic writing · Note-taking systems · Note-taking tools

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1 Overview

A note—the outcome or product of note-taking—has been associated with the notion of information unit (Siegel, 2018) or a knowledge building block that is stored externally (Forte, 2020d). Some consider notes to be private, unfinished and meaningful mainly to the note taker (Boch & Piolat, 2005; Forte, 2018a; Palmer et al., 2009; Siegel, 2018), whereas others have maintained that quality notes are understandable for someone unfamiliar with the content (Williams & Eggert, 2002). Note-taking, or note-making (Marin et al., 2021), serves two basic functions. The first—encoding—function entails recording information in the form of a note in order to commit information to memory, whereas the other—external storage—function concerns the use of notes as a form of external memory repository that may be accessed at any time (Jiang et al., 2018). Notes can be recorded with the use of a variety of analogue and digital tools. While analogue notes can be produced in paper notebooks, on pieces of paper, printed texts, post-its, whiteboards, flipcharts, etc., notes taken digitally require the use of electronic devices (Ahrens, 2017; Forte, 2018a; Kadavy, 2021; Marin et al., 2021) and are enabled through a keyboard, digital ink or voice (Khan et al., 2020). Notes are generated through a range of strategies, i.e. by underlining or highlighting a sentence or an excerpt or witing a comment in the margins (which is associated with what is known as text annotation), as well as summarising or verbatim transcription (of texts, lectures, videos), taking screenshots, voice memoing, mapping, sketching, outlining, and jotting down of ideas (Forte, 2018a; Friedman, 2014; Palmer et al., 2009). Consequently, notes may exist in various forms, such as a quote, a passage from a book or article, a summary, a bullet-point list, a photo, a drawing, a sketch, a voice memo, etc., in analogue and digital formats.

Some guidance on note-taking techniques (methods) is available. In general, these can be classified as linear and nonlinear; while linear note-taking resembles conventional text writing, non-linear note-taking often involves graphical representation of information (Makany et al., 2009). Recommendations and advice on note-taking techniques (methods) have been published to support L1 and L2 students' note-taking in lectures and classrooms, while reading and for written assignments (e.g., Hamp-Lyons, 1983; Lowen & Metzger, 2019; Sheridan, 2021; Siegel, 2016). Existing techniques can be classified according to their purpose. As described on the Sheridan website, in order to take notes in the classroom or during the lecture, students (note takers) may adopt the Cornell method, the outline method, the matrix method, concept mapping, PowerPoint slides, or use note-taking software. To take notes from reading, students may make use of the highlighting method, information funnel method and the SQ4R method. Finally, in order to take notes for written assignments, students may take advantage of the visualising method or the cue card method.

Digital (electronic) note-taking began with the use of a computer to type notes in a word processor, which then evolved into the use of various note-taking applications (Bennett & McKain, 2018). Evernote, created by Stepan Pachikov, is considered to be a pioneer among digital note-taking tools; its beta version was released in 2008, raising considerable interest among users and soon finding its followers (*The*

History of Evernote…, 2021). Although not very widely adopted in practice, the free Zettlkasten software based on Lhumann's system developed by Daniel Luedecke (http://zettelkasten.danielluedecke.de/en/) is yet another important contribution in the area of note-taking technology. Currently, note-taking applications that draw on the idea of bi-directional linking are gaining in popularity (Appleton, n.d.). In particular, Roam Research, founded by Conor White-Sullivan, and their knowledge management tool under the same name (Holm, Rowley, & Nisay, n.d.) have attracted a lot of attention (e.g., Daniels, 2020; Haisfield, 2020a, 2020b; Keiffenheim, 2021).

Digital note-taking has been increasingly considered part of a broader strategy of personal knowledge management, wherein diverse information is recorded, managed, and used for writing, for example. At the moment, two systems are gaining traction among those that deal with writing: the digital Zettelkasten system and Second Brain.

The Zettelkasten system, developed by Niklas Luhmann, was originally based on paper index cards containing references, short notes on ideas and the content of the literature, stored in wooden slip-boxes. Ahrens (2017), who adopted Luhmann's system to better serve modern-day writers, prescribes work towards the completion of a text in eights steps. The first four of these steps focus specifically on note-taking and involve making fleeting notes (capturing ideas connected to the writing project as they appear throughout the day), making literature notes (summarising the content), which is followed by making permanent notes (one's own elaboration of others' ideas), and then adding the permanent notes to the slip-box by storing them behind related notes. With the use of digital tools, this system, referred to as the digital Zettelkasten, has become even more powerful (Kadavy, 2021).

Second Brain—created by Tiago Forte, not a scholar of writing, but who has had an impact on how people take and use notes—is a personal knowledge management system for capturing, organising, and sharing knowledge using digital notes (Forte, 2019). Forte's CODE Framework (standing for Capture, Organise, Distill, Express) is a four-stage methodology for the management of knowledge that is saved in notes (Forte, 2019, 2020c). Specifically,

- capturing (or collecting)—involves saving the ideas and insights that one considers to be worthy of saving (e.g., webpages, PDFs, stories, book and conversation notes, excerpts of texts, quotes, images, screenshots, examples, statistics, metaphors, mindmaps);
- organising—entails organising content by current projects rather than by topics;
- distilling—involves condensing notes to a summary, and also connecting and organising them;
- expressing—entails creating the output in the form of a blog post, a YouTube video or a self-published ebook.

Theoretically, digital note-taking strongly connects to writing through the perspectives on writing that emphasise memory systems (Bereiter & Scardamalia, 2009; Hayes, 1996; Hayes & Flower, 1980) as well as self-regulation and self-efficacy

(Zimmerman & Risemberg, 1997) in the production of the written text. These theories allow to understand the function of note-taking in academic writing and the importance of regulatory processes at the junction of reading sources, note-taking and writing activities.

2 Core Idea of the Technology

In general, digital note-taking technology is built on the principles that highlight augmenting the capacity of human memory/mind, productivity enhancement, and collaboration with others in the creative process (Holm, Rowley, & Nisay, n.d.; *The History of Evernote…*, 2021). As to writing, digital note-taking eases the process in two main ways; first, by making note (idea) capturing and management more efficient and, second, by supporting writers' thinking. While processing information from books, articles, social media, webpages, etc., one's own and others' ideas can be instantly and easily saved with the use of a digital note-taking tool (Perell, 2021). Once taken, unlike analogue notes that can be lost, unintelligible, scattered and hard to be found when needed, digital notes have the advantage of being located in one place, which makes them more durable, searchable, accessible, shareable with others, and editable whenever a need arises (Forte, 2018b). In addition, writers can rely on the capacity of technology to store valuable ideas and resources long term, which frees writers from the limitations of their own memory (Forte, 2019). Finally, with the increasing body of saved ideas and emerging connections between them that become visible to the writer-note taker, note-taking technology supports writers' thinking and creativity and also prevents writer's block (Forte, 2019, 2020b; Perell, 2021).

However, there are no guidelines prescribing the use of note-taking tools. As these tools resemble a clean paper notebook, it is necessary to find a way to strategically work with a specific tool by, for example, adopting or developing a broader note-taking system (Forte, 2018a), such as the digital Zettelkasten system or Second Brain. Moreover, there is no predefined target user for a specific tool. In order to help users choose a suitable note-taking application, Duffy (2021) evaluates a range of tools and offers recommendations for various users, e.g., "best for business use and collaboration," "best for students," "best for creatives" (Duffy, 2021) or for specific purposes, e.g., "best for free and open source option," "best for organising with a small number of notebooks," "best for speed," "best for text notes only," "best for team notes and task management combined." Advice like this can help writers choose the tool that best fits their academic, professional, and personal needs.

3 Functional Specifications

Digital note-taking tools typically enable the following:

- *Capturing notes.* Notes can be typed (or handwritten) or imported in many ways, such as with the use of browser extensions for saving web pages, email capture, document scanning, third-party integrations, and by attaching files (audio notes, images, text documents, etc.).
- *Storage.* Notes can be autosaved in a note-taking application.
- *Mono-directional linking.* Notes can include links to external webpages as well as other notes within the application.
- *Text editing.* Text editors usually allow bolding, italics, bullets, numbering, they also offer different font styles, colours, and text sizes.
- *Tagging.* Tags added to a note help make connections between individual notes.
- *Organisation.* Notes can be arranged into pages, notebooks, folders, groups of folders (stacks), etc., that are devoted to, e.g., different writing projects.
- *Search.* Notes can be searched within the body of saved notes by keywords or tags.
- *Sharing notes with others.* Links can be created to share with, e.g., colleagues collaborating on writing projects, who will be able to see the note without needing to create an account.

Less common features include, among others:

- *Side-by-side viewing.* The setup of display windows in the note-taking application can enable the simultaneous viewing of two or more notes (pages).
- *Bi-directional linking.* Once saved in a note-taking tool, every time a note is mentioned in other notes, the original note automatically receives a link to the page the note is referenced to.
- *Filtering.* Unlike searching, note filtering allows to avoid viewing irrelevant information.
- *Running queries.* Notes can be searched in a more advanced way by using tags, page references and logical operators (e.g., AND) to extract entries of interest.
- *Templates.* Templates can be created to be reused for notes of a similar format, such as source metadata.

Importantly, the adoption of digital note-taking tools may be initially challenging as it takes time to take advantage of their potential and, consequently, the benefits of digital note-taking may not be immediately visible. In addition, the emergence of new note-taking applications may tempt users and exporting content between the applications may take their time and effort.

4 Main Products

There exist numerous digital note-taking tools compatible with different operating systems that allow for capturing and management of large amounts of notes. Comparisons and lists of (fast developing) software are available, such as https://en.wikipedia.org/wiki/Comparison_of_note-taking_software or https://en.wikipedia.org/wiki/Category:Free_note-taking_software, but, at the time of this writing, these present two major gaps. First, bi-directional linking tools (e.g., Roam Research, Craft, Obsidian, RemNote, Hypernotes, etc.) have not been included yet and, second, information about free plans is hardly available. With this in mind, Table 1 displays author-compiled comparison of selected tools in terms of pricing and operating systems, including bi-directional linking tools ($N = 24$).

5 Research

Note-taking for academic writing is not a well-understood area. Research has been conducted across various disciplines, such as information science, educational psychology, linguistics, education, including language education.

Research provides some initial insight into the types and purpose of notes taken by academic writers. Specifically, Qian et al. (2020) revealed that study participants—PhD students—produce three main kinds of notes while working towards a synthesis: in-source annotations (i.e. notes within a source), per-source summaries (written condensations of main information in a source) and cross-source syntheses (depictions of an overall grasp of the study problem emerging across the sources). In-source annotations were made to identify and record elements and observations that are relevant to one's research by highlighting on either printed or digital documents and using the comment function in PDFs. Peer-source summaries contained results, theories, concepts, solutions, as well as questions inspired by the paper, written in the form of a section or a paragraph, saved by some participants in reference managers, such as Mendeley or Zotero. In addition, it was found that a tagging system was used to collect papers that could be used in specific writing projects. Participants also mentioned using the image capture tools to take snapshots from a source (e.g., to capture the picture of a specific formula). Finally, cross-source syntheses took the form of outline summaries of key ideas across the sources or a mind map, among others.

There is also research demonstrating how the note-taking process assists scholarly writing. In particular, Qian et al. (2020) show how study participants progress from in-source annotations to per source summaries, and sometimes from per-source summaries to a synthesis, which is usually a non-linear process. Participants regarded the transition from annotations to cross-source synthesis as non-linear since their per-source summaries were not always adequate, necessitating a return to previous in-source annotations. Other research shows how language changes from source text

Table 1 Comparison of selected note-taking tools (Author's elaboration)

Application	Price plans*	Compatibility*
Agenda Notes	Free, premium features	iOS, macOS
Bear	Free features, pro	iOS, macOS
Craft	Personal (free), professional, team (coming soon)	iOS, macOS
Dynalist	Free, pro	Android, iOS, Linux, macOS, Windows
Evernote	Free, personal, professional and teams plans	Android, iOS, macOS, Windows
GoodNotes	$7.99	iOS, macOS
Google Drive	Free 15 GB of storage, 100 GB costs $1.99 per month, a terabyte costs $9.99 a month	Android, browsers, Linux, macOS, Windows
Google Keep	Free	Android, browsers, iOS
Hypernotes	Personal (free), plus, business, enterprise	Android, iOS, Linux, macOS, Snapcraft, Windows
Joplin	Basic, pro, business	Android, iOS, Linux, macOS, Windows
Metmoji Note Lite	Free	Android, iOS, Windows
Microsoft OneNote	Free	Android, browsers, iOS, Mac, Windows
Milanote	Free	Android, iOS, Mac, Windows
Notability	Free starter, paid subscription	iOS, macOS
Notepad +	Free, pro	iOS
Notion	personal (free), personal (pro), team and enterprise plans	Android, iOS, macOS, Windows
Obsidian	Personal (free), catalyst, commercial	Android, iOS, Linux, macOS, Windows
Penultimate	Free, in-app purchases	iOS
RemNote	Free, pro, lifelong learning	Android, iOS, Linux, macOS, Windows
Roam Research	Free 31-day trial, pro and believer plans, scholarships for "for scholars lacking financial stability"	Linux, macOS, Windows
Simplenote	Free	Android, browsers, iOS, Linux, macOS, Windows

(continued)

Table 1 (continued)

Application	Price plans*	Compatibility*
Standard Notes	Free, core, plus, pro	Android, browsers, iOS, Linux, macOS, Windows
Supernotes	Starter (free), unlimited, lifetime	Linux, macOS, Windows
Zoho Notebook	Free	Android, iOS, Linux, macOS, Windows

*as of 11th November 2021

to notes to a summary text (Hood, 2008). Still others revealed that collaborative note-taking does not improve academic writing performance, though this process leads to the retention of more information among the students (Fanguy et al., 2021), which suggests that social interaction may affect the outcomes of note-taking in different ways.

Literature also contains studies revealing the positive effect notes have on the quality of writing. Research has shown that any type of note-taking activity is related to the good quality of written expression (Lahtinen et al., 1997). In addition, notes may have a favourable effect on the language of student essays (Slotte & Lonka, 2001) and that note-taking may serve as a writing framework, especially if notes contained phrases allowing students to express themselves in their essays (Wilson, 1999).

The issue of how specific note-taking tools (apps) are used for writing has rarely been addressed. Qian et al. (2020) found that study participants used many tools which were different for each type of notes they took. For in-source annotations, the participants used printed paper, in addition to MacOSX Preview, Zotero and Mendeley reference managers, Google Drive, and PDF readers. For per-source summaries, the participants used Mendeley and Zotero, a paper notebook, MS OneNote, Google Doc, XMind mind mapping, PDF reader and Overleaf Latex Editor. For cross-source syntheses, they used MS Word (and/or online), Google Doc, MS OneNote, MacOSX Note, Mendeley, MS PowerPoint, mind/concept mapping, and Scrivener. The participants changed tools, using 4.2 tools on average, for taking different types of notes. Moreover, tools were adjusted to fit needs for which they were not originally designed, e.g., participants used reference managers to write per-source summaries.

Literature dealing with note-taking for academic writing contains very little information about the usability of note-taking tools. Importantly, as revealed by Qian et al. (2020), the necessity to switch between several tools and platforms, referred to as "tool separation" (p. 10) was a major source of friction for academics while note-taking and writing was regarded as disturbing or delaying the writing process. Some participants developed strategies to cope with these difficulties, e.g., displaying the reading window on the left and the note-taking window on the right at the same time. Research conducted among higher education students outside the area of academic writing shows that Evernote can be helpful for organising information (Kani, 2017) and can be more advantageous as lab notebook than a paper notebook (Walsh & Cho, 2013), but also reveals challenges (Kerr et al., 2015), such as these related to web clipping and file uploading (Roy et al., 2016). Literature also contains research reports on the design and implementation of programmes, such as a system for note-taking in lectures (Kam et al., 2005), a system for note-taking while watching online videos or a mobile application for collaborative note-taking (Petko et al., 2019).

6 Implications of Digital Note-taking Technology for Writing Theory and Practice

6.1 Digitalisation of Writing Spaces

Digital note-taking considerably expands the space for writing, affording writing in real (offline) and digital writing environments. With digital note-taking tools, academic writers can write anytime, whatever they need to, wherever they are, at any stage of text composition.

6.2 Digitalisation of the Writing Process

Digital note-taking is uniquely suited to address the demands of academic writing. Although there is no one prescribed use of digital note-taking tools for organising the writing process, existing note-taking systems can help writers by providing the methodology for the external storage and management of their notes with the outlook towards individual and collaborative writing. As such, note-taking systems implemented with the use of digital tools have the potential to make the complex task of writing easier and to alleviate the anxiety associated with writing.

6.3 Learning to Write Writing to Learn

Note-taking applications have the potential to support learning to write by allowing easy access to models for writing (in the form of saved texts) and modelling the process of writing. This may be afforded with the use of templates that can be created to scaffold less experienced writers in their writing, by, e.g., displaying instructions concerning what content is needed in specific parts of the text, how to structure and organise the text at the micro, meso and macro level, and what language can be used to meet particular rhetoric purposes.

6.4 Formulation Support

Notes can be a point of departure for the writing of one's own text when they include language from the sources read.

6.5 Competences Needed for Future Writers/Technological Knowledge

As academic writers deal with an abundance of information and ideas for writing, they are in need of developing competence around digital note-taking in order to build their own workflows. With this in mind, they need knowledge and skills to do with knowledge management, familiarity with note-taking systems, strategies for writing, as well as note-taking tools. In addition, writers would benefit from the mindset that recognises the value of systematic note-taking, and that is characterised by curiosity, creativity, enthusiasm for work with their notes for learning, thinking and, ultimately, writing.

6.6 Writing and Thinking

Digital note-taking tools support the cognitive processes envisioned in Bloom's revised taxonomy (Anderson & Krathwohl, 2001), i.e. remembering, understanding, applying, analysing, evaluating and creating. In particular, note-taking tools with bi-directional linking are of special significance as they help writers organise their thoughts or consider problems/phenomena from different perspectives. In so doing, these tools support writers' creativity and the development of original ideas. In addition, as individual differences in the capacity of writers' working memory, the ability to store and retrieve information form long-term memory, as well as self-regulation processes and self-beliefs may affect the writing process, digital note-taking tools have the potential to compensate for the limitations of human cognitive abilities by serving as a reliable external space, allowing to retain, retrieve, and manipulate a large amount of information.

6.7 Impact on Digital Writing Quality

Note-taking tools may positively influence the quality of writing in terms of text content, its organisation and language. As to content, note-taking tools afford the space for idea interaction and the emergence of unique insights that can find their way in the text. Concerning text organisation, note-taking tools may store models of the written genres in the form of, e.g., published articles, that writers can follow while writing own texts. In addition, writers may benefit from generated templates that can guide writers in the construction of whole texts, sections and/or paragraphs. With regard to language, saved texts (whole or excerpts) can also help writers to model their own language in allegiance with the genre requirements in a specific discipline or field.

6.8 Writing Opportunities, Assignments, and Genres

Digitally saved notes become a rich repository of ideas and resources for writing that can be readily drawn from in order to complete any writing task, in any genre.

6.9 Collaborative Writing and Collective Papers

Individual writers may have a huge store of information saved in their note-taking applications which can be shared with colleagues in the process of writing collective papers. In addition to this, writers can develop notes together. This leverages the opportunities for creativity and idea generation for collective writing.

6.10 Author Identities, Roles, and Audiences

With an efficient digital note-taking system, author identities as knowledge/information workers are likely to be positively affected. By depending on an external system to draw ideas or resources, they are more likely to be more able to regulate their writing and improve their self-beliefs. The role of an efficient note taker emerges as being of crucial importance for academic writers, serving a supporting function to help them learn and develop. With increased productivity and creativity, writers are likely to write more, publish more, and broaden their audience.

6.11 Feedback, Discussion and Support

Communities built around specific note-taking systems and/or note-taking tools may be a source of considerable support around digital note-taking. They can constitute a valuable venue for discussion and feedback, sharing of good practice, troubleshooting and serving as inspiration for others.

7 Conclusions and Recommendations

7.1 Writing Practices

Thoughtful application of note-taking systems with digital tools for knowledge management is particularly important in academic writing work and academic success as they may support writers' memory, creativity, self-regulation and self-efficacy. Writers can benefit from adopting note-taking systems and applications for smooth workflow in one central place, helping them to remember information, but also to think and write, individually and collaboratively. Hence, although mastering specific note-taking systems and tools may be time-consuming, this investment is needed to master the ability to efficiently take notes in order to write quality texts and be more productive. With the use of digital tools, academic writers need note-taking systems that support the writing process in terms of capturing inspirational insights, generating ideas, text drafting and editing, capturing models of genres, resources, managing references, generating in-text citations and bibliography.

7.2 Teaching

As note-taking is an acquired competence that exponentially improves writing, action is needed to incorporate insights from note-taking practice and research into academic writing pedagogy in order to help writers fulfil their potential. Teaching should aim to equip academic writers, both students and researchers, with the competences that allow them to effectively capture knowledge and manage saved notes needed for writing. Teaching could focus on developing note takers' abilities in the familiarity with and the use of whole note-taking systems, as well as digital tools for note-taking, bearing in mind note takers' writing goals, needs, interests and preferences.

7.3 Research

Research on note-taking for academic writing is very modest, more theoretical and empirical work is necessary as many areas require research attention. Most importantly, research is needed to define the concept of (digital) note-taking for academic writing. Furthermore, future work can conceptualise the products and processes of (digital) note-taking, it can also take cognitive processes and social and cultural context, as well as the role of digital mediation in note-taking practices into consideration. Importantly, note-taking research can benefit from insights from the literature concerning active reading and sense-making (cf. Qian et al., 2020) cognitive artefacts as extensions of human mind (cf. Heersmink, 2020), as well as semantic memory (cf. Kumar, 2021) and semantic memory networks (cf. Hills & Kenett, 2022; Kenett & Thompson-Schill, 2020) to better understand the notion of networked thinking underpinning note-taking for the creative process.

Empirical work is needed to better understand the products of note-taking in the context of academic writing, note-taking trajectories, obstacles and facilitators in the note-taking process. Research is also needed to explore the usability and effectiveness of digital note-taking systems and tools for academic writing, individual differences among writers note takers, the impact of digital note-taking on the quality of the written text written individually and collaboratively. The issue of language used to take notes should also receive more attention and linguistically-oriented research is called for to better understand the issues such as information unit, information flow, as well as meaning making while note-taking from sources to texts. It is also worth considering whether and how language proficiency affects L2 students' and researchers' ability to take notes when their written output in not in their mother tongue.

As to methodological recommendations, research is needed for in-depth analyses of a wide range of academic note takers, including students, beginning researchers, and expert scholars. Along with quantitative research methods (experiments and surveys), researchers can adopt qualitative methods (e.g., case studies, action research, ethnography) to describe good digital note-taking practices and analyse (un)successful academic note takers. Design-based research could be particularly beneficial in designing efficient note-taking products and processes by examining and eliminating encountered friction and difficulties.

7.4 Tool Development

Theoretical models (that are yet to come) and empirical findings (yet to be obtained) may inform the design and development of note-taking systems and tools to support note takers in academic writing contexts. System and tool development efforts need to recognise academic writers (students and researchers) as knowledge/information workers, their academic/research writing goals, information needs and challenges, and the specificity of writing processes as one of larger scholarly information practices/activities (Palmer et al., 2009), without forgetting that writing is only one of the aspects of academic worklife and that work is just one of the dimensions in an individual person's life, who may or may not capture knowledge in different types of notes and for different purposes over time. In particular, for the process of academic writing, friction between tools and products may be eliminated in order to assist academic writers manage their notes and resources. Tool designers could also consider incorporating guidance (e.g., text templates) in note-taking applications for novice academic writers as part of the note-taking system for capturing and/or creating notes at preliminary stages of text writing to facilitate writing.

8 List of Tools

Tool	Description	References
Agenda Notes; Agenda; https://agenda.com/	organising notes in the form of a timeline, assigning dates to notes, and connecting them to calendar events, etc.; free and premium features*; iOS, macOS	
Bear; Shiny Frog; https://bear.app/	linking notes and using hashtags to organise them, encrypting notes; includes a focus mode and advanced markup features, etc.; free features and pro plans*; iOS, macOS	
Craft; Luki Labs; www.craft.do	arranging similar ideas (notes) into subpages, creating cards, sharing notes and collaborating on documents, etc.; includes bi-directional linking; personal (free), professional, and team (coming soon) price plans*; iOS, macOS	
Dynalist; Dynalist; https://dynalist.io/	organising notes with the use of tags, cross linking, sorting, formatting and sharing notes, etc.; free and pro price plans*; Android, iOS, Linux, macOS, Windows	
Evernote; Evernote Corporation; https://evernote.com/	synching, organising and searching notes, creating tasks, to-dos, and templates; includes a web clipper, a document scanner, and a calendar; free, personal, professional and teams plans*; Android, iOS, macOS, Windows	Kani (2017); Kerr et al. (2015); Roy et al. (2016); Walsh and Cho (2013)
GoodNotes; GoodNotes Limited; https://www.goodnotes.com/	saving and managing both typed and handwritten notes, marking up PDFs and PowerPoint presentations, creating notebooks; enables side-by-side viewing, sharing documents, using flash cards, etc.; $7.99 price plan*; iOS, macOS	
Google Drive; Google; https://drive.google.com	storing notes in files and folders, allows to share and collaborate on documents; free 15 GB of storage, 100 GB costs $1.99 per month, a terabyte costs $9.99 a month*; Android, browsers, Linux, macOS, Windows	Qian et al. (2020)
Google Keep; Google; https://keep.google.com	saving and organising notes, photos and audio, creating lists, editing, sharing, and collaborating on notes, setting reminders about a note, automatic transcription of voice notes; free*; Android, browsers, iOS	
Hypernotes; Zenkit; https://zenkit.com/en/hypernotes/	saving and editing notes, bi-directional linking, structuring notes in a the form of a hierarchical outline, visualising concepts in a semantic graph, sharing and collaborating on notes, etc.; personal (free), plus, business, and enterprise price plans*; Android, iOS, Linux, macOS, Snapcraft, Windows	

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Tool	Description	References
Joplin; Laurent Cozic; https://joplinapp.org/	taking notes in the form of an image, video, PDF and audio file, publishing, sharing and collaborating on notes; includes a web clipper; basic, pro, and business price plans*; Android, iOS, Linux, macOS, Windows	
MetaMoJi Note Lite; MetaMoJi Corporation; http://noteanytime.com	taking and saving notes, enables drawing, sketching and annotating PDFs, among others; free*; Android, iOS, Windows	
Microsoft OneNote; Microsoft; https://www.microsoft.com/en-us/microsoft-365/onenote/digital-note-taking-app	saving and organising notes into sections and pages, navigating, searching, sharing and collaborating on notes; enables ink annotations; free*; Android, browsers, iOS, Mac, Windows	Qian et al. (2020)
Milanote; Milanote; https://milanote.com/	creating notes, uploading files, saving links, and organising notes in the form of boards; free*; Android, iOS, Mac, Windows	
Notability; Ginger Labs; https://notability.com/	creating, sharing and downloading notes from other users, marking notes with digital ink; free starter, paid subscription*; iOS, macOS	
Notepad + ; Apalon; https://www.apalon.com/notepad.html	taking and organising typed and handwritten notes, including sketches and drawings, annotating images; free and pro price plans*; iOS	
Notion; Notion Labs; https://www.notion.so/	capturing, editing and organising notes, sharing with others; personal (free), personal (pro), team, and enterprise price plans*; Android, iOS, macOS, Windows	
Obsidian; Obsidian; https://obsidian.md/	graph viewing, backlinking, daily notes, tagging, searching notes, recording voice notes, presenting notes as slides, and many more; personal (free), catalyst, and commercial price plans*; Android, iOS, Linux, macOS, Windows	
Penultimate; Evernote Corporation; https://evernote.com/products/penultimate/	taking handwritten notes and sketching, organising notes in notebooks by topic, project, etc., automatic syncing to Evernote; free, in-app purchases*; iOS	
RemNote; RemNote; https://www.remnote.com/	saving and linking notes, generating flashcards from notes, highlighting and referencing PDFs and web-based articles, etc.; free, pro, and lifelong learning price plans*; Android, iOS, Linux, macOS, Windows	
Roam Research (Roam); Roam Research; https://roamresearch.com/	bidirectional linking, formatting, searching and filtering notes, asl well as side-by-side, Kanban, and graph viewing, among many others; free 31-day trial, pro and believer plans, scholarships for "for scholars lacking financial stability"*; Linux, macOS, Windows	

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Tool	Description	References
Simplenote; Automattic; https://simplenote.com/	syncing, tagging, searching, backup, sharing, and publishing notes online; free*; Android, browsers, iOS, Linux, macOS, Windows	
Standard Notes; Standard Notes; https://standardnotes.com/	creating notes in nested folders; autocompleting tags, pinning notes, archiving notes, protecting notes with a passcode, etc.; free, core, plus, and pro price plans*; Android, browsers, iOS, Linux, macOS, Windows	
Supernotes; Supernotes; https://supernotes.app/	saving and searching notes, linking and nesting notecards, creating daily and thoughts collections, exporting notecards as PDF of Markdown files, sharing notecards; includes backlinking and a night mode; starter (free), unlimited, and lifetime price plans*; Linux, macOS, Windows	
Zoho Notebook; Zoho Corporation; https://www.zoho.com/notebook/	saving, tagging and organising notes in notebooks and stacks; protecting notes and notebooks with passcodes and Touch ID, sharing and collaborating on notes; free*; Android, iOS, Linux, macOS, Windows	

*As of 11th November 2021

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